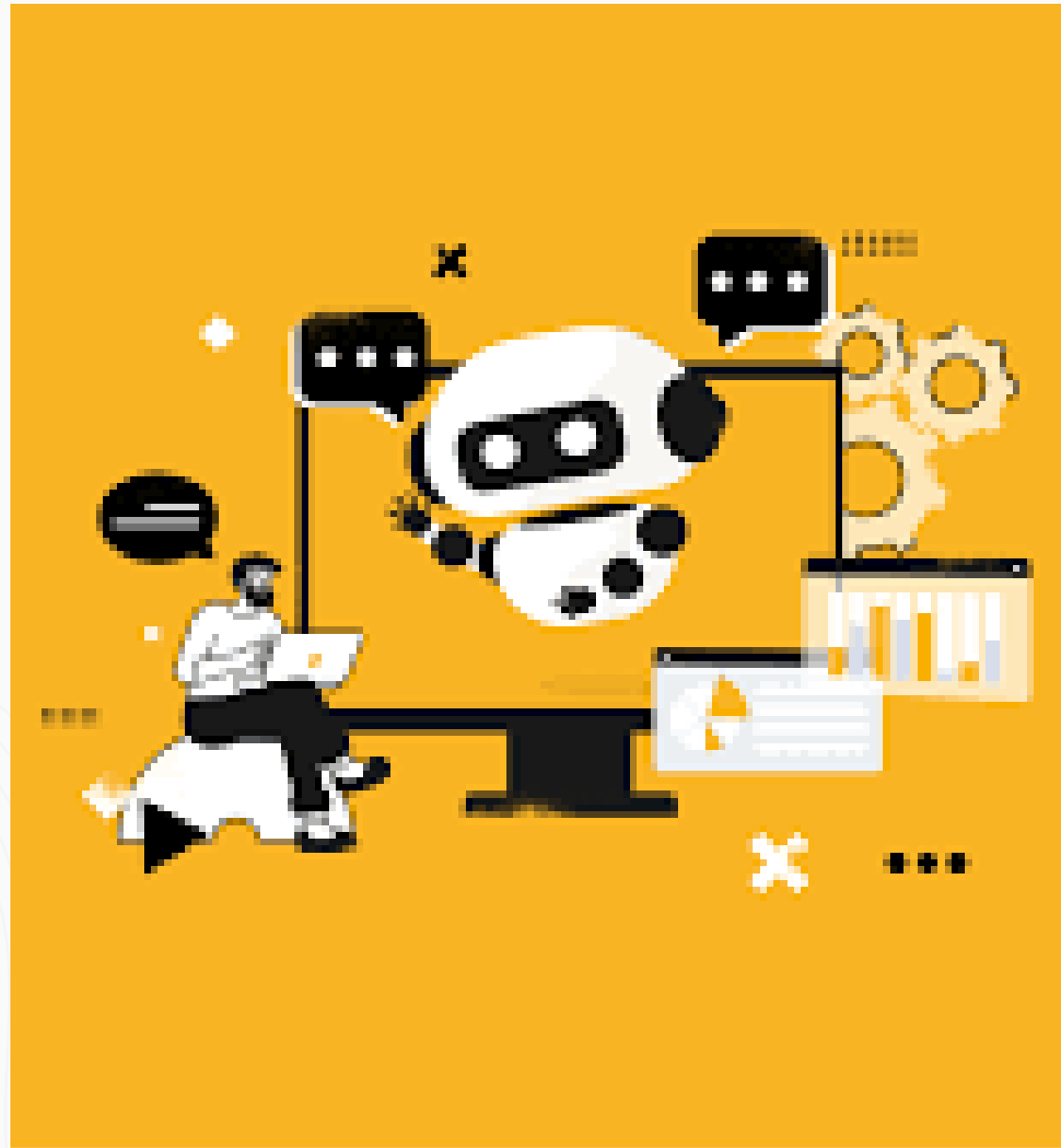




MFU Computer Engineering Smart Guide

Vibe Coding and RAG AI Project





Group Members

Group Number- 14

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Project Background

Prospective and current students need accessible information about the Computer Engineering program, including learning tracks, curriculum, career opportunities, fees, and contact details. This project supports faster academic information access through an AI-based chatbot.



Main Focus:

- Supports prospective and current students
- Provides quick access to program information
- Covers curriculum, careers, fees, and contact details
- Improves accessibility through an AI chatbot

Problem Statement

Computer Engineering program information is available, but users may need a more convenient way to access it based on their specific questions. This project addresses the need for a simple academic information system that can provide quick and relevant responses.

Main Issues:

- Different users need different types of program information
- Important details should be accessible in one place
- Answers should be clear, fast, and easy to understand



Project Objective



- Develop a chatbot for academic program information
 - Support prospective and current students
 - Provide quick answers to common program questions
 - Apply Vibe Coding and RAG AI in a practical project
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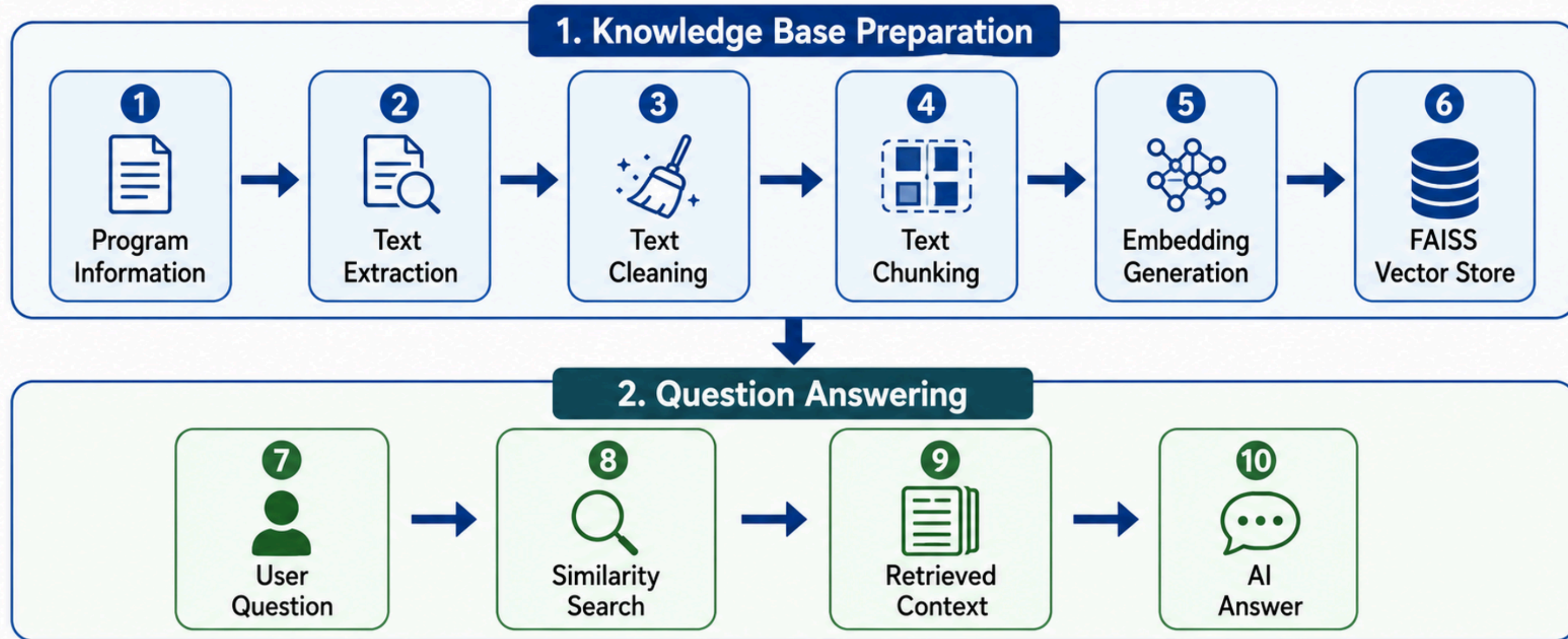
Dataset Description

We used the PDF file of Computer Engineering program information from the School of Applied Digital Technology, Mae Fah Luang University as Our Dataset

Main Information Covered
Program overview
Degree title
Specialized learning tracks
Career opportunities
Curriculum structure
Program Educational Objectives
Program Learning Outcomes
Tuition and fees
Further study opportunities
Contact information

RAG System Workflow

Retrieving relevant program information and generating answers



Purpose

This helps the chatbot answer questions using program-specific information instead of relying only on general AI knowledge.

Model and Tools Used

The project uses Python-based tools to build the RAG chatbot and deploy the application.

Tools and Libraries
Python
PyMuPDF for text extraction
Sentence Transformers for embeddings
BAAI/bge-small-en-v1.5 as embedding model
FAISS for vector search
Hugging Face model for answer generation
Joblib for saving model files
Streamlit for deployment
Gradio for Colab preview

Saved Files
best_ce_rag_model.pkl
scaler.pkl
app.py
requirements.txt

Key Features

Our chatbot helps users access Computer Engineering program information through a simple question-answering interface.

Main Features

- Retrieves relevant program information
 - Provides short and clear answers
 - Supports quick questions for common topics
 - Uses knowledge cards for important facts
 - Shows group number and group members
 - Supports Streamlit deployment
 - Supports Gradio preview in Colab
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Example Questions and Answers

Example 1

Question: What are the three learning tracks?

Answer: Hardware and Intelligent Control Systems, Data Science and Artificial Intelligence, and Data Communications and Networks.

Example 2

Question: How much is the total program fee?

Answer: The total program fee is 224,000 THB.

Example 3

Question: What career opportunities are available?

Answer: Computer Engineer, Data and AI Engineer, Software Developer, Network Engineer, Telecommunications Engineer, and related careers.

Conclusion

- This project applies Vibe Coding and RAG AI to develop an academic information chatbot for the MFU Computer Engineering program.
 - The chatbot supports prospective and current students by providing faster access to important program information.
 - Overall, the project demonstrates how RAG can be used to build a practical AI tool for academic information services.
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